Customer Support Agent Evaluation

Week 4 core track | Ramesh Suthan Palani | November 2, 2026

Goal and evaluation question

Route e-commerce tickets to the correct support team under a human-reviewed business policy. Does an explicit product-versus-refund priority rule improve classification while keeping response validity, latency and cost acceptable?

The agent is a single-node LangGraph classifier using Muse Spark Open on Fireworks AI. It returns one category and a short explanation. LangSmith records ticket-level inputs, outputs, reference IDs and nested model traces. Expected labels are not included in model messages.

Three controlled runs

Metric	Original 1024	Control 2048	Improved 2048
Accuracy	91%	96%	98%
Macro F1	0.927	0.962	0.990
Valid responses	96%	100%	98%
p95 latency (s)	16.86	11.65	13.21
Known cost (USD)	0.239355	0.259043	0.279014
Cost coverage	98%	100%	100%

Main finding: the same-settings prompt comparison improved accuracy from 96% to 98%. It fixed all four targeted routing errors, but two previously valid responses became invalid after reaching the output limit. This is a measured tradeoff, not a claim of production readiness.

All runs contain the same 100 cases. Errors remain in the denominator. The original run was resumed only for unattempted cases after provider failures; no failed result was replaced.

Dataset and reviewed routing policy

All 100 ticket texts came from the supplied CSV and were human reviewed before freezing. They comprise 25 related seed families. Four source labels (t004-t007) changed to refund_request. Descriptive tags add detail; they do not create new routing categories.

Category	Reviewed meaning
order_status	Tracking and delivery questions without an explicit non-delivery refund request.
refund_request	Non-delivery refunds, returns, cancellations, overcharges and refund follow-up.
product_issue	Damaged/wrong/misdescribed goods, damaged packaging and missing instructions. Damaged-product refund requests stay here.
account_help	Login/session issues, account security concerns, address and payment-method updates.
other	General shipping policy, feedback, stock, discounts and sale questions.

Scoring and pass bars

PASS means a valid output whose category exactly matches the reviewed label. Wrong categories, invalid responses and provider failures are FAIL. Accuracy uses all 100 cases; macro F1 averages the five categories. p95 latency uses linear interpolation. Costs use the saved Fireworks price snapshot and reported token usage; unknown usage is never treated as zero.

Metric	Target	Improved result
Accuracy	At least 95%	98% - met
Macro F1	At least 0.90	0.990 - met
Valid responses	100%	98% - not met
p95 latency	At most 15 seconds	13.21 seconds - met
Mean inference cost	At most $0.02 per case	$0.002790 - met

Original failure groups

4 damaged-product/refund priority errors (t044-t047); 1 packaging/manual scope error (t059); 2 output-limit failures (t015, t058); and 2 HTTP 500 provider failures (t026, t055). The first two groups concern prompt coverage; the latter groups concern response completion and infrastructure.

Prompt change and controlled comparison

First, the unchanged baseline prompt was rerun with max_tokens increased from 1024 to 2048. Then one focused priority clarification was added at 2048. Dataset, model, temperature zero, timeout and no-retry policy were unchanged. Packaging/manual wording was not expanded.

Original prompt

You route e-commerce customer support tickets into exactly one category.

order_status: questions about where an order is, tracking, or delivery ETA.

refund_request: the customer wants their money back.

product_issue: the item arrived broken, wrong, defective, or not as described.

account_help: login, password, address, or payment method changes.

other: general questions, feedback, browsing, or anything outside these categories.

Treat the next message as the customer's ticket. Return a JSON object with

"category" (one of the five exact keys) and "explanation" (one concise sentence).

Added instruction

When a ticket reports a broken, damaged, wrong, defective, or not-as-described product, choose product_issue even if the customer explicitly requests a refund or refuses a replacement. For a missing or undelivered order, an explicit request for money back is refund_request; without a refund request, it is order_status.

Observed fixes and regressions

Cases	Control 2048	Improved 2048
t044-t047	refund_request (wrong)	product_issue (correct)
t004-t007	refund_request (correct)	refund_request (correct)
t058	product_issue (correct)	Invalid response; output limit
t083	other (correct)	Invalid response; output limit

The prompt pair has 4 fixes, 2 regressions and 94 unchanged correct results. Both invalid responses used 2048 output tokens, including 2045 reported reasoning tokens. Their model traces show finish_reason=length. Increasing the limit did not eliminate this failure mode.

Trace evidence and repository guide

Control 2048 experiment

Improved 2048 experiment

course-t083: output-limit regression

course-t044: targeted routing fix

LangSmith links require access to the project workspace. The repository also contains saved ticket outputs, metadata and inspected trace evidence so the numerical findings can be reviewed without account access.

Artifact	Purpose
reports/customer_support_evaluation.xlsx	Source tabs plus Our Summary, Our Results, Our Reviewed Labels, Our Failure Analysis and Our Paired Results. Correct=1 means PASS; Correct=0 means FAIL.
notebooks/customer_support_evaluation.ipynb	Executed analysis notebook. Paid flags are off; saved-results replay makes no model calls.
data/frozen/v1 and artifacts/runs	Frozen labels, three original result sets, run specifications and metrics.
prompts and src/support_eval	Versioned prompt text and independently written agent/evaluation code.
reports/DEMO_SCRIPT.md	Narration and screen sequence for the remaining Loom recording.

Cost, limitations and attribution

Known project cost including the pilot is $0.79069535, excluding two original requests with unknown usage. Retaining both full-context reserves gives $3.42083935 conservative exposure under the approved $5 ceiling. These reserves are not measured charges.

The same synthetic set was used for diagnosis and reevaluation. Single runs cannot fully separate intervention effects from generation variability or provider conditions. Further work would test response completion on independent tickets. No extra experiment, LLM judge, deployment or external submission was performed.

Source tickets and baseline category descriptions: The Gen Academy, repository revision ef517b274cf86a2c4fa8ae72fdc89b3dce80f41d. The source workbook has 150 cases while the CSV has 100; only one shared ID has identical text. Supplied scores were not reused. This follows the Customer Support track; the separate Own Use Case checklist does not define this project's scope.

AI assistance was used for implementation, analysis and documentation. Ramesh made the ground-truth and routing-policy decisions and approved the experiment design. A presentation-style video with synthetic narration is included in reports/video. Loom hosting remains pending if required by the portal.